

Progress Report: Sprint 7

CS.028 Optimizing Research Software Codes - 26 January 2026

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Summary

- Discovered 6 bugs and created them as issues to solve in the next sprint.
- Solved 2 of the 6 bugs and made the program run 1.5 - 3 minutes faster in a 20-minute run.
- Continued researching into timing inconsistencies and csv parsing failures.
- Modulated parse() and calculate_positions() to isolate runtime anomalies.

Sprint Work Log

Fixed Conversion from Float to Datetime

Status: Complete

Author: Kathryn Butler

Reviewers: N/A

Source Link:

<https://github.com/KathrynJButler/optimizing-research-software-codes/commit/obofef7c3b5b658841bcd60470d4e47554c8ed4f>

Supporting Artifact:

```
39 - obs = xarray.Dataset({}, coords={"time": [], "sv": []})
39 + obs = xarray.Dataset({}, coords={"time": np.array([], dtype='datetime64[us]'), "sv": []})
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40 attrs: dict[str, T.Any] = {}
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41 for u in use:
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42     o = rinexsystem2(
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44     @@ -167,7 +167,7 @@ def rinexsystem2(
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```

Code diff in rinex2db.py - Datetime64 explicitly specified

Notes: After running into this issue multiple times during runtime tests, I noticed that often the program would throw an error about converting a float64 to a datetime64. I

fixed this by specifying that the datatype expected is a datetime object, so the program didn't immediately assume that the information it received was a float64.

Optimized Navigation Lookup Using Idxmin

Status: Complete

Author: Kathryn Butler

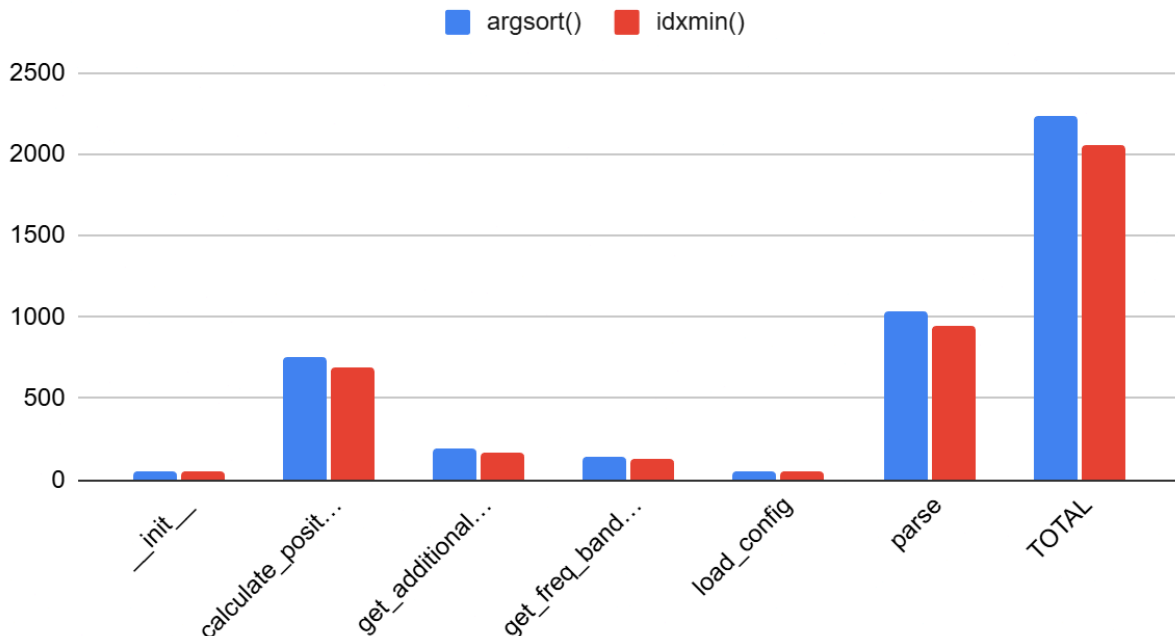
Reviewers: N/A

Source Link:

<https://github.com/KathrynJButler/optimizing-research-software-codes/commit/456977554a6462a3e75342d575c726e1f4158865>

Supporting Artifact:

argsort() vs idxmin()



Using `idxmin()` instead of `argsort()`, making things go measurably faster ([Source](#))

Notes: The initial error we were getting:

C:\Projects\optimizing-research-software-codes\gnss_python-main\rnx2db.py:
709: FutureWarning: Series.__getitem__ treating keys as positions is deprecated. In a future version, integer keys will always be treated as labels (consistent with DataFrame behavior). To access a value by position, use ``ser.iloc[pos]``

Line 709: `return nav.iloc[(nav['MJS'] - t).abs().argsort()[0]]`

From what I can tell, `nav['MJS'] - t` is the time difference between every navigation record and the target time `t`, `.abs()` converts all those values into positive ones, `.argsort()` sorts each argument, and `[0]` returns the smallest value (the closest time). This line can run close to 1.7 MILLION times for each run of the program, so it's really important to get it as efficient as possible.

There's a function that can find the smallest value without needing to sort the whole list over and over again called `idxmin()`. This finds the index of the smallest value and returns it, without needing to sort the array over and over again, saving time and memory. The new line 709 and 710 are:

```
idx = (nav['MJS'] - t).abs().idxmin()
return nav.loc[idx]
```

Decluttering the output is a big win, and it appears to have brought the average runtime down by 1.5-3 minutes in an originally 20-minute run.

Modulations of `parse()` and `calculate_positions()`

Status: Partial

Author: Joseph Schaab

Reviewers: N/A

Source Link:

https://github.com/KathrynJButler/optimizing-research-software-codes/blob/main/gnss_python-main/rnx2db.py

Supporting Artifact:

```
def parse(self) -> pd.DataFrame:
    """
    Parse observation and navigation file and calculate position, azimuth, and elevation
    """

    # Filter
    #self.obs = self.obs[
    #    (self.obs.C1.notna()) | (self.obs.C2.notna()) | (self.obs.C5.notna()) | (self.obs.C7.notna()) | (
    #        self.obs.C8.notna())
    #]

    #####
    # Break #
    #####
    tmp = []
    for c in self.constellations:
        tmp.append(self.obs[self.obs.index.get_level_values('sv').str.startswith(c)])
    self.obs = pd.concat(tmp)
    self.nav = self.nav[(self.nav.SVclockBias.notna())]

    #####
    # Break #
    #####
    # Add additional fields
    for field in ['PosX', 'PosY', 'PosZ', 'ClkCorr', 'FreqBand', 'SignalStrength', 'CarrierPhase', 'Doppler', 'CA', 'P', 'MJS']:
        self.obs.insert(self.obs.shape[1], field, np.nan)
    self.nav.insert(self.nav.shape[1], 'MJS', -1)
```

Notes: Documented modulation edits in OneNote:

https://oregonstateuniversity-my.sharepoint.com/personal/schaabj_oregonstate_edu/_layouts/15/Doc.aspx?sourcedoc={128ebaee-85b3-4cee-b3d1-948fe64d2cbb}&action=edit&wd=target%28Analysis.one%7C540b1621-cb09-474e-8b00-1b377594481f%2FDependencies%7C9c7fb113-9230-4b43-9a2e-f10df6e0d951%2F%29&wdorigin=NavigationUrl. Isolated subroutines within functions to analyze line-by-line runtimes. Will use debugger (such as gdb) to target lines of code that require updating.

Risk and Quality Tracking

Bug Count: 6 ([Link to issue query](#))

Top Bugs:

- [Treating Keys as Positions and Using Argsort](#)
 - **Severity:** High
 - **Owner:** Kathryn Butler
 - **Current Status:** Resolved
 - **Actions:** No further action needed. (Series.getitem treating keys as positions is deprecated. Look up how to solve it.)
- [Incorrect conversion from float to datetime](#)
 - **Severity:** Medium
 - **Owner:** Kathryn Butler
 - **Current Status:** Resolved
 - **Actions:** No further action needed. (Research into the line causing the problem. Determine the input causing the problem, and correct the datetime variable.)
- [Inconsistent timing data](#)
 - **Severity:** High
 - **Owner:** Kathryn Butler
 - **Current Status:** Ongoing
 - **Actions:** Create a smaller test data pool, run the tests multiple times to determine if the size of the file increases the likelihood of failed parsing attempts.
- [Xarray.merge conflict](#)
 - **Severity:** Medium
 - **Owner:** Kathryn Butler
 - **Current Status:** Ongoing
 - **Actions:** Research what “compat” does, and either set it explicitly or use the new combine_kwarg_defaults.
- [Numpy DataFrame.swapaxes is deprecated](#)
 - **Severity:** Medium
 - **Owner:** Kathryn Butler

- **Current Status:** Ongoing
- **Actions:** Research into the numpy process and determine if this bug affects runtime. If so, attempt to update the numpy version. Additionally, research into how using the most up-to-date libraries affect runtime/output consistency.
- [Log.txt hardcoded](#)
 - **Severity:** Medium
 - **Owner:** Kathryn Butler
 - **Current Status:** Ongoing
 - **Actions:** Log.txt seems to be announcing a preset number of cores to calculate. Figure out why it is set to only be 8 cores. Find out if you can manually adjust the core count to be something other than just 8, then add the text to the function dynamically.

Next Sprint Plan

- Create a smaller sample test dataset for faster testing. Link: https://github.com/KathrynJButler/optimizing-research-software-codes/tree/main/Datasets/COVL_1sec
- Continue researching into timing inconsistencies and csv parsing failures. Link: <https://github.com/KathrynJButler/optimizing-research-software-codes/tree/main/documentation>
- Update documentation to improve the setup process. Link: https://oregonstateuniversity-my.sharepoint.com/personal/schaabj_oregonstate_edu/_layouts/15/Doc.aspx?sourcedoc={128ebaee-85b3-4cee-b3d1-948fe64d2cbb}&action=edit&wd=target%28Analysis.one%7C540b1621-cb09-474e-8b00-1b377594481f%2FDependencies%7C9c7fb113-9230-4b43-9a2e-f10df6e0d951%2F%29&wdorigin=NavigationUrl

Team Process Reflection

- Our team optimized runtime performance by swapping instances of argsort() with idxmin(). We were able to achieve optimized runtime performance by allowing this task to run in $O(n)$ time instead of $O(n^2)$. Our team also modulated parse() and calculate_positions() to isolate instances where runtime must be improved to produce desired results.
- Our team still struggles to effectively communicate and meet at designated times throughout the week. Introducing a team calendar and establishing multiple team check-ins each week is our plan towards improving this issue.
- For actionable improvements, we will update our team calendar each week and present the calendar to our TA and Dr. Park. All team members will be involved in this process to keep us all on the same page. We will agree on two check-in times each week to update individual progress.